7.1 Angles of Polygons

Name _____ Period ____ Date _____

Geometry 7.1 (30-minute plan) · Student sheet

I can…

- find the sum of the interior angle measures of a polygon.
- use interior and exterior angle sums to find unknown angle measures.
- find each interior and exterior angle of a regular polygon.

Part 1 · Notes: fill in the blanks

Triangles in a polygon (from the TV): from one vertex, draw every diagonal and count the triangles.

Number of sides	Triangles from one vertex	Sum of the interior angles
4		
5		
6		
n		

Polygon Interior Angles Theorem: The sum of the interior angle measures of a convex n-gon is _____.

Corollary: The interior angle measures of a quadrilateral add to _____°.

Polygon Exterior Angles Theorem: The exterior angles of a convex polygon, one at each vertex, add to _____°.

Regular n-gon: each interior angle = _____ each exterior angle = _____

Part 2 · Examples (work them with the class)

Example 1

Find the sum of the interior angle measures of a convex 12-gon.

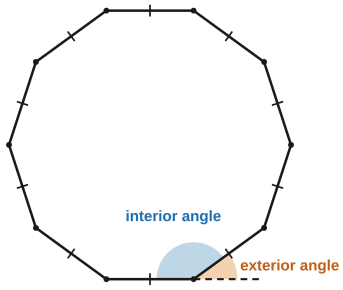
Example 2

A pentagon has interior angles of 110°, 2x°, 130°, (x + 25)°, and 95°. Find x and the two unknown angles.

Example 3

Find the measure of each interior angle and each exterior angle of a regular decagon.

regular decagon: 10 congruent sides, 10 congruent angles



Part 3 · Partner practice

Partner practice

Each exterior angle of a regular polygon measures 24°. How many sides does it have?